

# Jacob Dineen

AI, Reasoning & Cognition (ARC) Lab — ARIZONA STATE UNIVERSITY

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## RESEARCH INTERESTS

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I work on reasoning and alignment in large language models, with related interests in multi-agent reinforcement learning, controllability, and explainable AI (XAI). I have also held data science, ML engineering, and applied research roles across financial services and AI startups.

## EDUCATION

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| <b>ARIZONA STATE UNIVERSITY — Ph.D. Artificial Intelligence</b> <small>GPA 4.00/4.00 · Advisor: Ben Zhou</small> | 2022–2027 |
| <small>Committee: Muhao Chen, Chitta Baral, Vivek Gupta</small>                                                  |           |
| <b>UNIVERSITY OF VIRGINIA — M.Sc. Computer Science</b> <small>GPA 3.96/4.00 · Advisor: Madhav Marathe</small>    | 2019–2021 |
| <b>SYRACUSE UNIVERSITY — M.S. Data Science</b> <small>GPA 4.00/4.00</small>                                      | 2017–2018 |
| <b>GRAND CANYON UNIVERSITY — B.S. Finance &amp; Economics</b> <small>GPA 3.65/4.00</small>                       | 2012–2015 |

## RESEARCH EXPERIENCE

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|-----------------------------------------------------------------------------------------|--------------|
| <b>Graduate Research Assistant, ARC Lab (ARIZONA STATE UNIVERSITY)</b>                  | 2024–Present |
| • Research on reasoning, alignment, controllability, and multi-agent RL in LLMs.        |              |
| <b>Graduate Research Assistant, SEFCOM (ARIZONA STATE UNIVERSITY)</b>                   | 2022–2024    |
| • AI and cybersecurity research.                                                        |              |
| <b>Applied Research, CAPITAL ONE — Center for Machine Learning (C4ML)</b>               | 2020–2021    |
| • Built reinforcement learning and agent-based simulations for organizational dynamics. |              |
| <b>Research Assistant, BIOCOMPLEXITY INSTITUTE (UNIVERSITY OF VIRGINIA)</b>             | 2019–2020    |
| • Worked on graph dynamical systems, cooperative game theory, and behavioral modeling.  |              |

## PROFESSIONAL EXPERIENCE

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|-------------------------------------------------------------------------------------------|--------------|
| <b>AI/ML Engineer, ALUMBRA AI</b> <small>PART-TIME</small>                                | 2026         |
| <b>Research, PARETO AI</b> <small>PART-TIME</small>                                       | 2025–Present |
| <b>Machine Learning Engineer, SPRING OAKS CAPITAL</b> <small>FULL-TIME</small>            | 2022–2025    |
| <b>Data Scientist, CAPITAL ONE</b> <small>FULL-TIME</small>                               | 2020–2022    |
| <b>Analyst &amp; Business Intelligence, REAL WORLD MARKETING</b> <small>FULL-TIME</small> | 2016–2019    |
| <b>Data Scientist, BUFFALO CHECK LLC</b> <small>PART-TIME</small>                         | 2015–2019    |
| <b>Optimization Analyst, VOLTARI</b> <small>PART-TIME</small>                             | 2012–2015    |

## PUBLICATIONS

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### Google Scholar Profile

<sup>+</sup> Corresponding Author / Mentor

### Peer-Reviewed Conference Proceedings (C)

- C1. Ye, Xiao, Li, Zhaonan, **Dineen, Jacob**, Xu, Zhikun, Lu, Shijie, Shen, Ming, Shrivastava, Shaswat, Ahuja, Avneet, & Zhou, Ben (2026). *Learning Verifiable Reasoning Programs through Cohort Consistency*. In *Proceedings of the 2026 Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the International Joint Conference on Natural Language Processing (ACL-IJCNLP 2026)*.
- C2. Shen, Ming, Xu, Zhikun, Ye, Xiao, **Dineen, Jacob**, & Zhou, Ben (2026). *BOW: Training Language Models to Reason Over Plausible Next Words*. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*.
- C3. Xu, Zhikun, Feng, Yu, **Dineen, Jacob**, Shi, Taiwei, Zhao, Jieyu, & Zhou, Ben (2026). *Skill Reuse as Compression in Agentic RL*. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*.
- C4. **Dineen, Jacob**, RRV, Aswin, Xu, Zhikun, & Zhou, Ben (2026). *Vocabulary Dropout for Curriculum Diversity in LLM Co-Evolution*. In *Proceedings of the Conference on Language Modeling (COLM 2026)*.
- C5. **Dineen, Jacob**<sup>+</sup>, RRV, Aswin, Liu, Qin, Xu, Zhikun, Ye, Xiao, Shen, Ming, Li, Zhaonan, Lu, Shijie, Baral, Chitta, Chen, Muhao, & Zhou, Ben (2025). *QA-LIGN: Aligning LLMs through Constitutionally Decomposed QA*. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 20619–20642.
- C6. RRV, Aswin, **Dineen, Jacob**, Handa, Divij, Uddin, Md Nayem, Parmar, Mihir, Baral, Chitta, & Zhou, Ben (2025). *ThinkTuning: Instilling Cognitive Reflections without Distillation*. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 31236–31250.
- C7. Xu, Zhikun, Shen, Ming, **Dineen, Jacob**, Li, Zhaonan, Ye, Xiao, Lu, Shijie, RRV, Aswin, Baral, Chitta, & Zhou, Ben (2025). *ToW: Thoughts of Words Improve Reasoning in Large Language Models*. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 3057–3075.
- C8. **Dineen, Jacob**<sup>+</sup>, Kridel, Donald J., Dolk, Daniel R., & Castillo, David G. (2023). *Unified Explanations in Machine Learning Models: A Perturbation Approach*. In *Proceedings of the 56th Hawaii International Conference on System Sciences (HICSS-56)*, pp. 795–804.
- C9. **Dineen, Jacob**<sup>+</sup>, Haque, A. S. M. Ahsan-UI, & Bielskas, Matthew (2021a). *Formal Methods for an Iterated Volunteer's Dilemma*. In *Proceedings of the 14th International Conference on Social, Cultural, and Behavioral Modeling (SBP-BRiMS 2021)*, pp. 81–90.
- C10. **Dineen, Jacob**<sup>+</sup>, Haque, A. S. M. Ahsan-UI, & Bielskas, Matthew (2021b). *Reinforcement Learning for Data Poisoning on Graph Neural Networks*. In *Proceedings of the 14th International Conference on Social, Cultural, and Behavioral Modeling (SBP-BRiMS 2021)*, pp. 141–150.
- C11. Dolk, Daniel R., Kridel, Donald J., **Dineen, Jacob**<sup>+</sup>, & Castillo, David G. (2020). *Model Interpretation and Explainability towards Creating Transparency in Prediction Models*. In *Proceedings of the 53rd Hawaii International Conference on System Sciences (HICSS-53)*.

## Peer-Reviewed Workshop Papers (WS)

- WS1. Srinivasan, Adarsh, **Dineen, Jacob**<sup>+</sup>, Sarfraz, Muhammad Uzair, Afzal, Muhammad Umar, Riaz, Irbaz, Zhou, Ben (2026). *RECAP: Transparent Inference-Time Emotion Alignment for Medical Dialogue Systems*. In *Proceedings of the 8th Workshop on Clinical Natural Language Processing (Clinical NLP) @ LREC 2026*, pp. 350–368. ORAL
- WS2. Li, Zhaonan, Chickering, Kyle R., Li, Bangzheng, **Dineen, Jacob**, Ye, Xiao, Xu, Zhikun, Lu, Shijie, Huang, Yuxi, Shen, Ming, Nguyen, Bach, Pavuluri, Jaya Adithya, Nguyen, Mau Son, Chavan, Sanika, Le, Ngoc Minh Thu, Chen, Muhao, Zhou, Ben (2026). *VisAnalog: A Diagnostic Suite for Visual Concept Transfer on Natural Images*. *CVPR Workshop on Visual Concepts (VisCon)*, 2026. ORAL

## Working Papers / Under Review (W)

- W1. Li, Zhaonan, Lu, Shijie, Wang, Fei, **Dineen, Jacob**, Ye, Xiao, Xu, Zhikun, Liu, Siyi, Cho, Young Min, Li, Bangzheng, Chang, Daniel, Nguyen, Kenny, Yang, Qizheng, Chen, Muhao, Zhou, Ben (2025). *Unbiased Visual Reasoning with Controlled Visual Inputs*. ARXIV PREPRINT
- W2. Liu, Qin, **Dineen, Jacob**, Huang, Yuxi, Zhang, Sheng, Poon, Hoifung, Xiao, Chaowei, Zhou, Ben, Chen, Muhao (2025). *ArenaBench: Automatic Benchmark Evolution via Multi-Model Competitive Evaluation*. ARXIV PREPRINT
- W3. RRV, Aswin, **Dineen, Jacob**, Handa, Divij, Parmar, Mihir, Zhou, Ben, Baral, Chitta, Mishra, Swaroop (2026). *Mid-training with Self-Generated Data Improves Reinforcement Learning in Language Models*. UNDER REVIEW, NEURIPS 2026
- W4. Ye, Xiao, **Dineen, Jacob**, Zhu, Evan, Lu, Shijie, Song, Kevin, Zhou, Ben (2026). *HINDCAST: Replaying Prediction Markets to Evaluate LLM Forecasters*. ARXIV PREPRINT
- W5. Ye, Xiao, **Dineen, Jacob**, Li, Zhaonan, Xu, Zhikun, Chen, Weiyu, Lu, Shijie, Huang, Yuxi, Shen, Ming, Tran, Phu, Yum, Ji-Eun Irene, Khan, Muhammad Ali, Afzal, Muhammad Umar, Riaz, Irbaz Bin, Zhou, Ben (2025). *Evaluating Medical LLMs by Levels of Autonomy: A Survey Moving from Benchmarks to Applications*. ARXIV PREPRINT

## SKILLS

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|                     |                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LLM Methods</b>  | pre/mid/post-training, RL (GRPO, RLHF), reward modeling, self-play, agentic tool use, RAG, synthetic data generation, evaluation and benchmarking                       |
| <b>ML Libraries</b> | PyTorch, TensorFlow, JAX, Polars, Dask, HuggingFace, TRL, vLLM, VeRL, PySpark, NetworkX, DGL, Torch Geometric, BoTorch, SnowparkML, Anthropic/OpenAI SDKs               |
| <b>Languages</b>    | Python, Rust, C, C++, Java, JavaScript, R, Bash, PRISM, x86-64                                                                                                          |
| <b>Databases</b>    | MySQL, PostgreSQL, MongoDB, Snowflake, Redis, SQLite, Redshift                                                                                                          |
| <b>Tools</b>        | Git, GitHub Actions, Docker, Kubernetes, Helm, Airflow, AWS (ECS, ECR, S3, EKS, CodeBuild), SageMaker, Databricks, FastAPI, Next.js, pytest, OR-Tools, Sigma, Streamlit |

## SERVICE

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**Conference Reviewer:** HICSS, SBP-BRIMS, NAACL, EMNLP, EACL, COLM

**Teaching / Tutoring:**

- CGCC Calculus & Linear Algebra Tutor (2019)
- Teaching Assistant — ASU CSE 365 / pwn.college (Security)

## MISCELLANEOUS

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**Expert AI Trainer, PARETO AI** (2024–Present)

**Independent Contractor, MERCOR INTELLIGENCE** (2025–Present)

**Cybersecurity:** [pwn.college](https://pwn.college) green belt (binary exploitation), user: jdin

## GRADUATE COURSEWORK

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**ARIZONA STATE UNIVERSITY — PhD Computer Science:**

Knowledge Representation; Computer Systems Security; Software Security; Planning and Learning Methods in AI; Algorithms; Research/Dissertation Hours.

**UNIVERSITY OF VIRGINIA — MSc Computer Science:**

Algorithms; Machine Learning; Computer Vision; Formal Methods; Reinforcement Learning; Graph Mining; Learning Theory (Game Theory); Cloud Computing; Research Hours.

**SYRACUSE UNIVERSITY — MS Data Science:**

Data Analysis and Decision Making; Business Analytics; Financial Analytics; Marketing Analytics; Advanced Information Systems; Data Science; Data Warehousing; Text Mining; Scripting for Data Analysis; Information Policy.